Day-Ahead Flight Arrival-Delay Risk for NYC Departures: Leakage-Aware Features, Logistic Regression versus Gradient Boosting, and Calibration under Shift

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***Abstract*—This Phase 1 study ranks flights leaving New York (EWR, JFK, LGA; 2013) by the risk of arriving at least 15 minutes late, using only predictors available at 20:00 on the evening before departure. Outcome-availability cutoffs also govern fitting, cross-validation, calibration and threshold selection. Regularised logistic regression and XGBoost are compared with a base-rate prior and route-history reference. On the chronologically reserved Nov–Dec window, the CV-selected logistic model reaches PR-AUC 0.394, against 0.404 for XGBoost, 0.353 for route history and 0.258 for the prior. Flagging the riskiest 10% each day gives precision 0.445 and lift 1.72. The test bootstrap does not establish a difference between the learned models. Static calibration fails in December (ECE 0.143); causal rolling recalibration improves probability quality in this run. Test exposure during development is disclosed. The offline analysis measures review-candidate ranking; schedule changes and delay savings are not simulated. Deep learning is reserved for a later phase.**

***Index Terms*—flight delay prediction, data leakage, gradient boosting, probability calibration, temporal validation, decision support**

Key Findings

1) **Leakage is measurable.** 31.1% of the previous day's arrivals are unknown at 20:00; moving the cutoff to midnight inflates CV PR-AUC by ≈0.010. Fitting cutoffs exclude 167 training and 251 calibration outcomes unavailable at the respective deployment boundaries.
2) **Both models beat both references in observed AP.** Test PR-AUC: LogReg 0.394, XGBoost 0.404, route history 0.353, prior 0.258.
3) **Model differences remain uncertain.** CV 0.4715 vs 0.4474; test-difference CI $[-0.013, +0.037]$ includes zero. Logistic regression is selected by the declared CV rule; this is not a demonstration of equivalence.
4) **Static calibration breaks under season shift.** Test ECE 0.082 (static Platt) vs 0.027 (rolling Platt); December ECE 0.143.
5) **A capacity-limited desk gains from ranking.** The top 10% per day is 1.72× more precise than random (Sec. X); this is ranking quality, not realised savings.

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I. Introduction

Flight delays cascade: a late inbound aircraft, a congested departure bank or an upstream weather system turns one disruption into many. Airlines, airports and passengers all benefit from knowing *in advance* which departures are most at risk, because the remedies (adding buffer to a rotation, swapping an aircraft, re-timing a departure, pre-positioning staff) take time to execute. This report studies that day-ahead ranking problem on the public `nycflights13` data, with the emphasis on *doing it honestly*: every feature must be computable at the moment the prediction is issued, every choice must be justified without looking at the test period, and every probability must be checked for calibration.

A. Problem and motivation

- **Task.** For each flight scheduled to depart EWR, JFK or LGA on day D, estimate $P(\text{arrival delay} \geq 15 \text{ min})$ at **20:00 NYC time on** $D-1$, and rank the flights by that estimate.
- **Why a decision clock matters.** Every past outcome is present in the table, so a careless join, rolling window or lag silently lets the future into the features. A high score obtained that way answers a question no operator can ask in real life. The 20:00 clock fixes exactly which outcomes are usable.
- **Why probabilities and not only ranks.** A desk that triages by rank needs only the ordering, but a desk that sets a threshold, budgets buffer time or compares airlines needs *calibrated* probabilities. We therefore evaluate both ranking (PR-AUC, ROC-AUC, lift) and probability quality (Brier, ECE).
- **Why this data.** `nycflights13` covers one full year at three airports, which gives strong seasonality (calm autumn, extreme December). That makes a chronological split a built-in stress test of calibration under shift.

B. Contributions

- A **decision clock**: an outcome enters a feature or a fitting/selection stage only once it is available at that stage's cutoff; leakage is quantified and unit-tested (Sec. IV).
- A **controlled model comparison** of logistic regression and XGBoost [1] against two references, with the final model fixed by a rule declared before test.
- A study of **calibration under seasonal shift** [2], [3]: a pre-declared static rule and a causal rolling alternative.
- A **policy-level decision-support analysis** for a capacity-limited review desk (Sec. X).
- A **development protocol** that discloses which test numbers were seen, and when, and keeps selection evidence separate from exploratory findings.

C. Scope

This is the machine-learning evaluation phase of Project 21, *Flight Delay Prediction and Rescheduling*, whose full brief is titled *Flight Delay Prediction via Gradient Boosting and Temporal Deep Learning*. Phase 1 covers literature review, data quality and EDA, feature engineering, theoretical justification, model application, reproducible code and an IEEE report supporting presentation and viva.

- **Implemented:** logistic regression and XGBoost on schedule, carrier, route, airport congestion, calendar and past-delay features; temporal evaluation, calibration and offline review-candidate ranking.
- **Later-phase DL:** a compact MLP with learned airline, origin and destination embeddings. A temporal model is optional and requires correctly ordered histories; both must use the same clock and date splits as the ML models.
- **Current limits:** no archived weather forecasts, cost model or simulated schedule changes (Sec. XI-D). Observed weather for D is unavailable at 20:00 on $D-1$. The full brief's what-if rescheduling deliverable remains incomplete beyond candidate ranking.

TABLE I
Notation.

Symbol	Meaning
D	target (departure) day
$a(t)$	availability date of an outcome known at time t
$y \in \{0,1\}$	late-arrival label (arr_delay ≥ 15 min)
$m,\ w$	shrinkage strength, history window (days)
C	inverse L2 strength of logistic regression
$\hat{r}_{k,D}$	shrunk delay rate of entity k for day D
$g_i,\ h_i$	gradient and Hessian of the loss for sample i
$k\%$	daily review capacity (share of flights flagged)

D. Organisation

Section II reviews related work and states the research gap. Sec. III–V describe the data, decision clock and features. Sec. VI gives the models and calibration theory, Sec. VII the evaluation protocol. Sec. VIII reports the evidence behind every decision and Sec. IX the exploratory test results. Sec. X is the decision-support analysis, Sec. XI discusses limits and validity, and Sec. XII concludes.

E. Notation

Table I lists the symbols used throughout.

II. Related Work

This section places the work in six strands and ends with the questions motivating the implementation. Bibliographic records identify the cited works; the comparison matrix marks details not assessed in the available sources rather than inventing feature sets or performance claims.

A. Flight-delay prediction

- **Surveys.** Sternberg et al. [4] organise the literature by scope, data and computational method, and highlight the growing use of machine learning. This is the map against which our design (single airport group, one year, tabular model) can be placed.

- **Classical machine learning with explicit horizons.** Rebollo and Balakrishnan [5] study random-forest departure-delay forecasting with explicit horizons and local/network delay information. The horizon is part of the task definition, and recent network state motivates our day-level history group. We do not transfer their task-specific errors or claim directly comparable performance for our arrival-delay classifier.
- **Scalable data mining and deep learning.** Belcastro et al. [6] study scalable data mining for flight delays, and Yu et al. [7] study a deep-learning approach to commercial-flight delays. Their full feature sets, calibration and validation protocols were not assessed here; the matrix states this explicitly. The Yu et al. bibliographic record was verified against Crossref during this revision.

B. Gradient boosting and tabular data

- **Boosting.** Friedman [8] casts boosting as steepest descent in function space; XGBoost [1] adds second-order optimisation, explicit regularisation and a fast tree learner. We use it as the flexible learner against logistic regression.
- **Trees versus deep nets on tabular data.** Grinsztajn et al. [9] find that tree ensembles remain state of the art on medium-sized tabular data. That supports XGBoost as a serious Phase 1 baseline before any deep model.

C. Leakage

- **Formulation.** Kaufman et al. [10] define leakage as the introduction of information about the target that should not legitimately be available at prediction time, and discuss avoiding it through a clear *learn–predict separation*. Our decision clock is an operational version of that separation for a daily forecasting problem: availability dates enforce which outcomes may feed which prediction.

D. Probability calibration and shift

- **Post-hoc calibrators.** Platt scaling [11] fits a sigmoid to classifier scores; isotonic-regression-based calibration [12], [13] fits a monotone step function and is more flexible but more data-hungry; beta calibration [14] generalises the sigmoid with a richer map family. Guo et al. [2] popularise ECE and temperature scaling for neural networks.
- **Calibration under shift.** Ovadia et al. [3] evaluate uncertainty under benchmark shifts; Rabanser et al. [15] study shift detection, including domain classifiers; Gama et al. [16] survey concept-drift adaptation. These motivate checking deployment probability quality and distribution change, but they do not establish how a calibrator will perform on this flight dataset.
- **Scoring rules.** The Brier score [17] and log-loss are strictly proper scoring rules [18]: they are optimised only by the true probability, which is why we optimise log-loss and report Brier.

E. Imbalanced evaluation

- **PR versus ROC.** Davis and Goadrich [19] relate PR and ROC curves and show that optimising ROC area does not optimise PR area; Saito and Rehmsmeier [20] show PR plots are more informative on imbalanced data. Our base rate is about one quarter late, so PR-AUC is the primary ranking metric and ROC-AUC is reported alongside.

F. Temporal validation and uncertainty

- **Rolling origin.** Tashman [21] reviews out-of-sample forecasting tests; Bergmeir and Benítez [22] discuss how temporal dependence affects cross-validation. We use forward rolling-origin CV inside the training window so that each validation period follows its fitting data.
- **Block bootstrap.** Künsch [23] introduces resampling *blocks* for dependent observations. Flights on one day are correlated, so we resample whole days (a block of one day).

G. Comparison matrix

Tables II, III, IV and V group the cited work by task and method. "–" means not applicable; "not assessed" identifies details not established in this review. The tables provide design context, not a leaderboard of comparable flight-delay scores.

H. Research gap

- **Operational availability.** This implementation must audit which predictors and fitting labels are known at the chosen day-ahead cutoff, applying learn–predict separation [10] to the complete pipeline.
- **Probability quality under seasonal change.** We compare a static calibrator and a causal rolling alternative across the observed seasonal change, motivated by calibration-under-shift studies [3].
- **Unresolved representation gap.** Manual route, airport and congestion summaries may miss interactions between time, carrier, airport and recent delay dynamics. Later embedding-based MLP experiments will test this hypothesis; the present model comparison does not prove that a DL model will improve performance.
- **Operational interpretation.** Daily top-$k\%$ ranking evaluates a review-capacity assumption. Intervention feasibility and simulated delay reductions require a separate what-if analysis.

These statements are limited to the papers we could confirm; we make no claim about the literature as a whole.

III. Data and Exploratory Analysis

We first audit the data, fix the target, and then read the exploratory patterns as design constraints. Table VI at the end of the section maps each observation to a design decision.

A. Data quality

- **Source.** The project uses the recommended Kaggle Flights CSV [24], derived from `nycflights13` [25]: 336,776 departures from EWR, JFK and LGA in 2013, with identifiers, carrier, route, distance, schedule and

TABLE II
RELATED WORK, FLIGHT-DELAY AND BOOSTING PAPERS (1/2): DATA, HORIZON, FEATURES AND MODEL.

Paper	Year	Dataset	Horizon / decision time	Weather features	Model
Rebollo & Balakrishnan [5]	2014	US air-traffic delay data	Explicit forecast horizons	not assessed	Random forests
Belcastro et al. [6]	2016	Flight records	not assessed	not assessed	Scalable data mining
Yu et al. [7]	2019	Commercial flights	not assessed	not assessed	Deep learning
Sternberg et al. [4]	2017	– (review)	–	–	Taxonomy of methods
Friedman [8]	2001	– (method)	–	–	Gradient boosting machine
Chen & Guestrin [1]	2016	Several benchmarks	–	–	Gradient-boosted trees (XGBoost)
Grinsztajn et al. [9]	2022	Tabular benchmark (∼45 sets)	–	–	Trees vs deep nets
This work	2026	nycflights13 (2013)	20:00 on $D-1$	None (by design)	LogReg, XGBoost

TABLE III
RELATED WORK, FLIGHT-DELAY AND BOOSTING PAPERS (2/2): METRICS, CALIBRATION, TEMPORAL HANDLING AND LIMITATION.

Paper	Metrics	Calibration reported	Leakage / temporal handling	Limitation (for our purpose)
Rebollo & Balakrishnan	Task-specific; not compared here	not assessed	Forecast horizon explicit	Departure-delay task differs from ours
Belcastro et al.	not assessed	not assessed	not assessed	Full protocol not assessed
Yu et al.	not assessed	not assessed	not assessed	Full protocol not assessed; DL is later-phase
Sternberg et al.	–	–	–	Survey; no new model
Friedman	Loss-specific fit	No	–	Not calibrated by design
Chen & Guestrin	Benchmark scores	No	–	Probabilities not guaranteed calibrated
Grinsztajn et al.	Benchmark accuracy	No	–	Not flight data
This work	PR-AUC, ROC-AUC, Brier, ECE, lift	Yes, static and rolling	Decision clock + leakage tests	Single year, no weather

TABLE IV
RELATED WORK, CALIBRATION, SHIFT, EVALUATION AND LEAKAGE PAPERS (1/2): DATA, HORIZON AND METHOD.

Paper	Year	Dataset	Horizon / decision time	Method
Brier [17]	1950	Weather forecasts	–	Brier score
Platt [11]	1999	SVM outputs	–	Sigmoid calibration
Zadrozny & Elkan [12]	2002	Benchmark classifiers	–	Calibration of classifier scores
Niculescu-Mizil & Caruana [13]	2005	UCI-style benchmarks	–	Many learners + Platt / isotonic
Kull et al. [14]	2017	Benchmark classifiers	–	Beta calibration
Guo et al. [2]	2017	Image / text benchmarks	–	Neural nets + temperature scaling
Ovadia et al. [3]	2019	Shifted benchmarks	–	Deep nets, uncertainty methods
Rabanser et al. [15]	2019	Benchmark shifts	–	Shift detection methods
Gama et al. [16]	2014	– (survey)	–	Drift adaptation
Gneiting & Raftery [18]	2007	– (theory)	–	Proper scoring rules
Davis & Goadrich [19]	2006	–	–	PR / ROC theory
Saito & Rehmsmeier [20]	2015	Simulated and real	–	PR vs ROC study
Tashman [21]	2000	– (review)	–	Out-of-sample tests
Bergmeir & Benítez [22]	2012	Time series	–	Blocked cross-validation
Künsch [23]	1989	Stationary series	–	Block bootstrap
Kaufman et al. [10]	2012	– (framework)	–	Leakage formulation and detection
This work	2026	nycflights13 (2013)	20:00 on $D-1$	LogReg/XGBoost, Platt/isotonic, temporal evaluation

recorded outcomes. The raw input is stored locally as `data/raw/flights.csv`.

- **Integrity checks (all passed).** No duplicate flights by identifier or by (date, carrier, flight, origin, scheduled time); the departure delay equals the clock-time difference modulo 24 h for every flight; hour and minute match the scheduled departure for every flight; no non-positive air time.
- **One anomaly.** EWR–EGE and JFK–EGE each have two distances (1726/1725 and 1747/1746) that change by exactly one mile on 1 March 2013: a dated reference-data

TABLE V
RELATED WORK, CALIBRATION, SHIFT, EVALUATION AND LEAKAGE PAPERS (2/2): METRICS, CALIBRATION, TEMPORAL HANDLING AND LIMITATION.

Paper	Metrics	Calibration reported	Leakage / temporal handling	Limitation (for our purpose)
Brier	Mean squared probability error	Yes (score)	–	Aggregate score only
Platt	Probabilistic outputs	Yes	i.i.d. calibration set	Static fit
Zadrozny & Elkan	Calibration quality	Yes	i.i.d.	Static fit
Niculescu-Mizil & Caruana	Calibration, ranking	Yes	i.i.d. splits	Static calibration
Kull et al.	Calibration quality	Yes	i.i.d.	Static; not temporal
Guo et al.	ECE, NLL	Yes (ECE)	i.i.d. test sets	No temporal shift
Ovadia et al.	Accuracy, NLL, ECE	Yes	Explicit dataset shift	Deep nets; not delay data
Rabanser et al.	Detection power	–	Explicit shift	Detection, not calibration
Gama et al.	–	–	Drift handling	Survey; no calibration focus
Gneiting & Raftery	–	Theory	–	Theory only
Davis & Goadrich	PR, ROC	–	–	Theory only
Saito & Rehmsmeier	PR vs ROC	–	–	Not a delay study
Tashman	–	–	Rolling origin	Forecasting, not classification
Bergmeir & Benítez	Forecast error	–	Blocked CV	Time-series regression
Künsch	–	–	Dependent data	Statistical theory
Kaufman et al.	–	–	Leakage theory	General; not flight-specific
This work	PR-AUC, ROC-AUC, Brier, ECE, lift	Yes, static and rolling	Decision clock + leakage tests	Single year, no weather

update with no practical effect, so we keep the values.

- **Outcome missingness is structural.** The code identifies 8,255 cancellations from missing departure time and uses departed flights with missing arrival delay (1,175) as a diversion proxy. These are dataset-derived definitions, not independently verified operational flags. All 2,512 missing tail numbers belong to cancelled flights. Missing arrival-delay labels are excluded from the main task rather than imputed. Separately, the logistic preprocessing pipeline median-imputes missing numeric predictors and adds missingness indicators, fitting these transformations only on eligible training rows.

B. Target

- **Label.** $y = \mathbb{1}[\text{arr_delay} \geq 15]$, using the BTS 15-minute late-arrival convention [26]. The positive rate is 24.5% of completed flights (16.1% at 30 min, 8.7% at 60 min).
- **Why arrival, not departure delay.** The arrival delay is what passengers and downstream connections experience; a flight that leaves late but makes up time is not a late arrival.
- **Cancelled and diverted flights.** They have no arrival delay, so they are dropped from the target; but they stay in the *schedule* counts, because a cancellation is not known the evening before. A sensitivity analysis counts them as delayed instead (Sec. IX).
- **Consequence for modelling.** A moderate imbalance ($\sim$1 in 4): PR-AUC is the main ranking metric, and class weights are not used so that predicted probabilities stay interpretable.

C. Patterns that drive the design

- **Time of day.** The delay rate rises from 10.5% at 05:00 to 37.4% at 21:00 (Fig. 1) because delay propagates through the day. $\Rightarrow$ cyclic time-of-day features plus the raw hour.
- **Season.** Monthly rates range from 13.7% (September) to 33.5% (December). September–October are calm and December is extreme, so a chronological split is a built-in stress test for calibration; with a single year, month effects cannot be separated from one-off events.
- **Carrier.** Rates range from 12.9% (HA) to 38.8% (F9) and some carriers are tiny (OO: 29 flights). $\Rightarrow$ one-hot encoding with a minimum frequency, and shrinkage of history rates.
- **Memory.** The lag-1 autocorrelation of the daily delay rate is 0.40 and falls to about 0.12 over lags 8–14: disruption is bursty with short memory. $\Rightarrow$ day-level history features alongside entity-level ones.
- **Congestion and the time-of-day confound.** Busier banks have more delays, but busy banks cluster in the evening where delays are high anyway. Splitting banks into terciles *within each departure hour* holds the hour fixed (Fig. 2): the busiest third is still 4.5 percentage points later than the quietest, and later in 13 of 16 hours. Part of the raw effect is time of day, but not all of it.
- **Multicollinearity.** The history rates are strongly intercorrelated (nested windows on the same key; network and origin rates that move together), some at $|r|$ close to 1 (Fig. 3). $\Rightarrow$ logistic regression must be regularised, and single-feature importance is misleading, so we use *grouped* permutation importance.

IV. DECISION CLOCK AND LEAKAGE

Leakage is information about the target that should not be available when the prediction is made [10]. Here it can arise through post-departure predictor columns, historical aggregates using outcomes before they are known, or fitting and selecting a model with labels unavailable when that model is needed. The clock must govern all three paths.

TABLE VI
FROM EXPLORATORY OBSERVATION TO DESIGN DECISION.

Observation	Design decision
Structural outcome missingness; cancellation/diversion proxies	Drop missing labels, keep schedule counts; impute missing numeric predictors within fitting folds
Strong time-of-day gradient (10% to 37%)	Cyclic departure and arrival time plus the raw hour
One year; seasonal months; calm Sep–Oct, extreme December	No month feature; chronological split; calibration studied under shift
Short-memory daily shocks (lag-1 autocorrelation ≈0.4)	One-day and seven-day lagged rates (day-level history group)
Stable carrier, route and aircraft propensities; some tiny groups	7- and 14-day entity rates with Bayesian shrinkage (CV-tuned m, window); one-hot with minimum frequency 50
Congestion correlates with delay but is confounded with the hour	Scheduled-bank counts, kept and tested by ablation
Leakage risk from post-departure columns and unavailable labels	Apply the 20:00 clock to features, model fitting, CV, calibration and thresholds; regression tests cover each boundary
History rates strongly inter-correlated	L2-regularised logistic regression; grouped, not per-feature, permutation importance

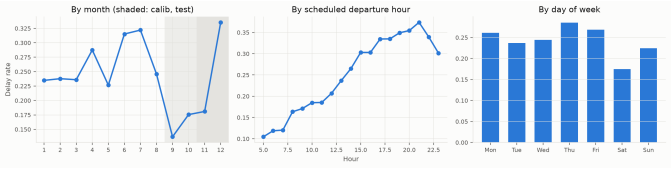


Fig. 1. Delay rate by month, hour and weekday.

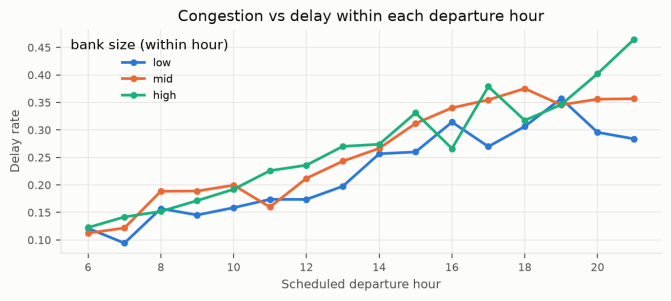


Fig. 2. Delay rate by scheduled-bank size, holding the departure hour fixed.

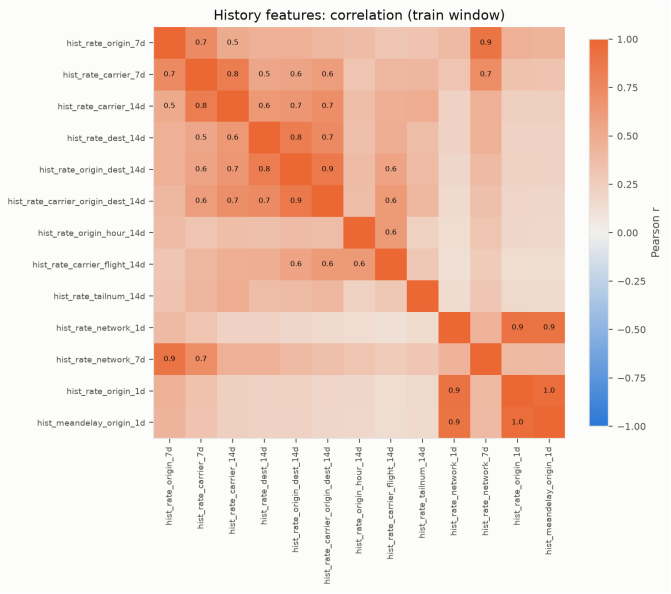


Fig. 3. Correlation of the history features (training window only).

A. Definition

- **Decision time.** 20:00 NYC on $D-1$ for every flight of day D.
- **Known when.** An arrival delay is known at the *actual gate arrival*, converted to the NYC clock. Cancellation status is assumed known at scheduled departure for cancelled flights, or actual departure for flights that leave; the dataset has no cancellation-notification timestamp.
- **Availability date.** The first target day whose decision time is at or after the known-time t:

$$a(t) = \lceil t - 20\,\mathrm{h} \rceil_{\mathrm{day}} + 1 \text{ day}.$$

- **Windows roll over availability dates, not flight dates.** A flight dated $D-1$ that lands at 22:30 has $a(t) = D+1$ and cannot help predict D. The timetable of day D (including flights that will be cancelled) is allowed; anything that happens on D is not.

B. Fitting and selection cutoffs

For a stage needed on target date D_0, a label is eligible only if $a(t) \le D_0$; unknown availability times are excluded. This is distinct from assigning the flight itself to a chronological split.

- **CV fitting.** Each fold uses earlier flight dates and only labels available at the first validation day's $D-1$ clock. The May, June, July and August fitting sets contain 99,160, 127,268, 154,318 and 182,626 rows; respectively 272, 292, 317 and 302 otherwise chronological training labels are excluded.
- **CV selection.** Validation labels used to select the model must be available before its first calibration prediction on 1 September. This also excludes 167 late-August validation outcomes.
- **Final fitting.** Of 211,684 training scoring rows, 211,517 labels are available by 20:00 on 31 August; 167 are

TABLE VII
DESTINATION OFFSETS RELATIVE TO NYC IN THE LOOKUP TABLE.

Offset (min)	Destinations
0 (Eastern)	60
−60 (Central)	21
−120 (Mountain)	8
−180 (Pacific, incl. PHX)	14
−240 (ANC)	1
−360 (HNL)	1

excluded from the final models and fitted prior.

- **Static calibration and thresholds.** September candidate fits exclude outcomes unavailable before 1 October. October checks, the final Sep–Oct refit and frozen threshold use outcomes available before 1 November: 55,377 labels, excluding 251. Later labels may still be scored retrospectively, but cannot determine these earlier fits.

C. Time zones and daylight saving

- **Problem.** `arr_time` is in the *destination's* local time and the dataset has no time-zone table.
- **Solution.** A **fixed lookup table** of 105 destination offsets relative to NYC (Table VII) plus the 2013 daylight-saving rule (10 March 02:00 to 3 November 02:00; HNL, PHX, SJU, BQN, PSE and STT do not observe DST). The arrival on the NYC clock is scheduled departure plus departure delay plus the local-time difference corrected for the offset.
- **Why not estimate offsets from the data.** Our first version did, and past features then depended on future data; the future-dependence test caught it. The table is static geography. It reproduces the data-derived offsets (checked in the notebook and by a test), and the arrival times it gives imply plausible taxi times for flights with an air time.

D. Size of the leak

- **Which outcomes are late.** 31.1% of $D-1$ arrivals land after 20:00 (usable only from $D+1$) and are delayed 38.1% of the time, against 18.3% for arrivals known by 20:00. The unknown outcomes are the *bad evening* the model should anticipate.
- **Clock sensitivity.** Moving the cutoff from 20:00 to midnight on $D-1$ admits later information than the stated task allows (Table XIX): CV PR-AUC rises from **0.4715** to **0.4814**, about **0.010**. These sensitivity runs apply the alternate clock to fitting as well as features; the difference is an empirical clock comparison, not a clean estimate of feature leakage alone.
- **Why not simply lag two days.** A 00:00 cutoff on $D-1$ gives 0.4589, lower by 0.013 on average and in all four folds, discarding the many $D-1$ outcomes already available by 20:00. The chosen clock retains usable recent information.

E. Leakage tests

- **(a) Future outcomes.** Overwrite every post-departure column from 15 June onward with random values and rebuild all features. Changed feature cells up to and including the cutoff: **0** (bit-for-bit identical); after the cutoff: 4,558,344, so the test is not vacuous.
- **(b) Decision clock.** For a target day D, randomise the arrival outcomes of $D-1$ flights that landed after 20:00. Changed cells on day D for 14 Feb, 11 Jul and 20 Dec: **0**.
- **Positive control.** Randomising the $D-1$ flights that landed *before* 20:00 changes 12,428, 13,078 and 12,740 cells; a test that cannot fail proves nothing, so this control confirms the check is sensitive.
- **(c) Fitting boundaries.** Regression tests change unavailable labels and verify invariant fitted LogReg/XGBoost predictions, prior, calibration choices and frozen thresholds. Additional tests cover equality at the cutoff, unknown times, CV eligibility and cancellation/diversion timing. Metric tests also cover constant, tied and unique probabilities. The complete suite passes 42 tests.

TABLE VIII
FEATURE GROUPS (n = NUMBER OF FEATURES).

Group	n	Content
Identifiers	3	carrier, origin, destination
Schedule	10	cyclic time-of-day and weekday, departure hour, weekend, days to holiday (cap 14), log distance
Congestion	7	scheduled flights in the origin-hour, ±1 h bank and day, destination and carrier volumes, leg index
Entity history	18	smoothed delay rate and log-volume by route, carrier-route, carrier, destination, origin×hour, flight number, tail number
Day-level history	7	network and per-origin rates over 1/7 availability days, cancellation rates, clipped mean delay

V. FEATURES

A. Feature groups

There are 45 features in five groups (Table VIII). Groups matter beyond bookkeeping: ablation and permutation importance operate on whole groups, because correlated members of one group can cover for each other.

- **Schedule.** Time of day and weekday are encoded as sine/cosine pairs so that 23:59 and 00:01 are neighbours; the raw hour is kept as well. Days to the nearest US federal holiday are capped at 14 days.
- **Congestion.** Counts of *scheduled* flights in the same origin-hour and neighbouring banks; cancellations are not subtracted, because they are unknown the evening before. The leg index (the aircraft's nth departure of the day) is a partial proxy for rotation.
- **Entity history.** Smoothed late rates and log-volumes for entities that carry persistent delay propensity: routes, carriers, destinations, origin-hours, flight numbers and tail numbers.
- **Day-level history.** Network-wide and per-origin rates over one- and seven-day windows of availability dates: the recent disruption state as far as the clock allows. A one-day

availability window is not a literal interval of actual arrival times.

B. Shrinkage of entity rates

- **The problem.** A tail number seen once and late has an empirical rate of 100%; a raw rate from a handful of flights is extremely noisy.
- **Beta-binomial shrinkage.** Treat the entity's late probability as having a Beta prior centred on the network rate π_D with strength m pseudo-flights; the posterior mean is

$$\hat{r}_{k,D} = \frac{\text{late}_k + m\,\pi_D}{\text{flights}_k + m},$$

where the counts cover outcomes *available* for D within the window and π_D is the network delay rate over the 28 days before D (also past-only).

- **Behaviour.** With few flights the estimate stays near π_D; with many, it approaches the entity's own rate. m sets how many flights it takes to trust the entity.
- **Worked example.** EWR→CAE is a thin route with one evening flight every three to four days; every flight lands after 20:00, so an outcome is usable only from two days later. With one prior late flight and $\pi_D = 0.247$, $m = 5$ gives $(1 + 5 \times 0.247)/6 = 0.372$, not 1.0.

C. Joint tuning of m, window and C

- **Why joint.** Hand-picked values ($m = 20$, 28 days) were replaced by a rolling-CV search. The best C depends on how noisy the features are, so each (m, w) pair is scored at its own best C of the logistic grid; fixing C could mislead the choice.
- **Search.** $m \in \{5, 20, 50\}$, window $w \in \{14, 28, 56\}$ days, nine pairs (Table XIX).
- **Result.** Less shrinkage is better at every window, and the 14-day window is best at every m. The CV-best triple is $m = 5$, $w = 14$, $C = 0.1$ (0.4715). The fitting-cutoff correction retains this triple.
- **How large is the gain.** Small relative to the fold spread (the old hand-picked pair, $m = 20$, $w = 28$, now scores 0.4665 with its own best C). The pipeline checks that the configured pair agrees with the CV-best pair.

D. Design choices

- **No month features.** Training covers months 1–8 and each seasonal month is observed only once. Adding cyclic month features lowers logistic CV AP from 0.4715 to 0.4691 and fixed-tree XGBoost from 0.4474 to 0.4405 (worse in two of four folds for each; Table XX). Trailing histories retain recent seasonal information. An earlier test-based justification was withdrawn (Sec. VII); this check uses the selected model settings without separately retuning them for month features.
- **Day-level history is its own group.** `hist_rate_network_1d` takes one value per day, shared by hundreds of flights, so its effective sample size is about the number of training days and a flexible model could use it as a date identifier. Keeping it separate lets the ablation test it in isolation.
- **Encoding.** LogReg one-hot encodes identifiers with minimum frequency 50; numeric predictors use median imputation, missing indicators and standardisation within each fitting pipeline. XGBoost uses native categorical splits and native numeric missing-value handling; it does not use the LogReg preprocessing pipeline.

VI. Models and Training

A. Logistic regression

- **Model.** $P(y = 1 \mid x) = \sigma(w^\top x + b)$, trained by minimising a convex L2-regularised log-loss (the intercept b is unpenalised):

$$\mathcal{L}(w) = -\tfrac{1}{n}\textstyle\sum_i \left[y_i \log p_i + (1 - y_i)\log(1 - p_i)\right] + \frac{1}{2Cn}\|w\|^2.$$

- **Why regularise.** The history features are strongly correlated, which makes unregularised coefficients unstable. The penalty is equivalent to a zero-mean Gaussian prior on w (MAP estimation); smaller C shrinks harder.
- **Why it is a strong baseline here.** The history features are already non-linear summaries (smoothed rates), so a linear model on top can capture most of the signal, and it is fast, convex and interpretable (coefficients, grouped permutation importance).
- **Choosing C.** From $\{3{\times}10^{-4}, 10^{-3}, 3{\times}10^{-3}, 10^{-2}, 0.1, 1\}$ by rolling CV (Table XXI, Fig. 12); the curve is flat above $C \approx 0.003$ and the fold spread is far larger than the difference between values, so $C = 0.1$ is the marginal maximum, not a sharp optimum.

B. XGBoost

- **Model.** An additive ensemble of trees $F_T(x) = \sum_t f_t(x)$ fitted stage-wise [1], [8]. At stage t it minimises a second-order Taylor approximation of the loss plus a complexity penalty:

$$\tilde{\mathcal{L}}^{(t)} = \textstyle\sum_i \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t(x_i)^2\right] + \gamma J_t + \frac{1}{2}\lambda\|v_t\|^2,$$

with $g_i = p_i - y_i$ and $h_i = p_i(1 - p_i)$ for log-loss, J_t the number of leaves in the new tree and v_t its leaf-weight vector. The implementation searches λ and leaves γ at its default zero.

- **Leaf weights and split gain.** For a fixed tree structure the optimal leaf weight is $w_j^* = -G_j/(H_j + \lambda)$, where G_j, H_j sum gradients and Hessians in the leaf. A split is accepted when

$$\text{Gain} = \tfrac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma > 0.$$

- **Role of the knobs.** *Depth* caps interaction order (depth 2 means pairwise interactions at most). *Minimum child weight* is a minimum Hessian mass per leaf, which guards against fitting tiny groups on a weak, noisy signal. *Column subsampling* and λ regularise; *learning rate* trades the number of trees against smoothness.
- **Search.** 48 combinations of depth $\{2, 3, 5\}$, minimum child weight $\{20, 100\}$, column subsample $\{0.5, 0.8\}$, $\lambda \in \{1, 10\}$ and learning rate $\{0.03, 0.1\}$ (row subsample fixed

at 0.8); 16 sampled at random, each with up to 3,000 trees and early stopping (100 rounds) on each fold's validation month. The tree count of a configuration is the mean of its four best iterations.

- **Selected.** Depth 2, learning rate 0.1, minimum child weight 20, row subsample 0.8, column subsample 0.5, $\lambda = 1$, **148 trees**; in-fold CV 0.4761 ± 0.0703. Its fixed-tree CV score is 0.4474. These settings come from the reproduced search; their selection does not by itself prove that the underlying signal is weak or smooth.
- **Why log-loss.** A strictly proper scoring rule [18]: its expectation is minimised only by the true probability, so training pushes toward calibration on the training distribution. The Brier score [17] is also proper; accuracy and AUC are not.

C. Reference models

- **Prior.** The eligible fitting-label base rate for every flight (PR-AUC equals the evaluated window's base rate; ROC-AUC is 0.5). It provides a reference with no flight-specific information.
- **Route history.** The smoothed 14-day carrier-route delay rate alone, with no model. A learned model must beat this simple, transparent heuristic to justify itself.
- **Placement.** Both appear in the principal model-comparison and operating-policy tables and relevant evaluation figures.

D. Calibration methods

- **Platt scaling** [11] fits $\sigma(a \operatorname{logit} p + b)$ on held-out scores. The positive fitted slope in this run preserves ranking. **Temperature scaling** [2] restricts this mapping to $b = 0$ and $a = 1/T$, $T > 0$; it is discussed but not implemented here.
- **Isotonic regression** [12], [13] fits any monotone step function by pool-adjacent-violators: more flexible, but it needs more data and can overfit.
- **Beta calibration** [14] is a richer parametric family; we mention it as an alternative and do not use it.
- **Shared assumption.** All of them assume the calibration data resemble deployment data. Sec. IX shows what happens when that fails.

VII. Evaluation Protocol

A. Split and cross-validation

- **Chronological split** (Table XXII): train Jan 8–Aug (211,684 rows), calibration Sep–Oct (55,628), test Nov–Dec (53,991). The first week is excluded as warm-up for the history windows.
- **Scoring versus fitting.** These are chronological scoring-row counts, not the eligible fitting counts in Sec. IV. Static train/calibration scores are retrospective diagnostics; they should not be presented as historical deployed predictions or independent estimates of generalisation.
- **Observed base-rate shift.** 26.5% in training, 15.7% in calibration, 25.8% in test (18.1% in November, 33.5% in December). These differences are properties of the data in the chronological windows, not a manipulated class balance.
- **Rolling-origin CV** inside the training window [21], [22]: train Jan–Apr $\rightarrow$ validate May; Jan–May $\rightarrow$ Jun; Jan–Jun $\rightarrow$ Jul; Jan–Jul $\rightarrow$ Aug. Each fold validates on a month strictly after its training data.
- **Abandoned design.** A single validation month (September) was dropped: it is the calmest month and its best XGBoost tree count differed from October's on nearly flat curves (Fig. 13).

B. Metrics

- **PR-AUC** is average precision, $AP = \sum_k (R_k - R_{k-1}) P_k$, a step-wise sum, not a trapezoidal area; the no-skill value is the base rate [19], [20]. **ROC-AUC** is reported alongside; it ignores the base rate.
- **Brier score** $= \frac{1}{n} \sum_i (p_i - y_i)^2$ [17], a proper score combining calibration and sharpness.
- **ECE** [2] is $\sum_b \frac{|B_b|}{n} |\bar{y}_b - \bar{p}_b|$ over probability bins. It depends on binning and sample size and is a marginal calibration check, not a measure of discrimination. Equal-width ECE uses 15 bins. Approximately equal-mass ECE starts with 15 balanced rank bins; cuts move to the end of tied groups and duplicate cuts collapse. Equal probabilities stay together, so bins may differ in size or number. Equal-width ECE and Brier remain the main diagnostics.
- **Lift** = policy precision / base rate.
- **Oracle recall@p.** The best recall at precision $\geq p$ with the threshold tuned *on the evaluated labels*. It is a point on the PR curve for comparing models, never an operating result.

C. Operating policies (fixed before test)

- **Primary: daily top 10%.** Flag $\lceil 0.10 n_D \rceil$ of each day's flights. This fixes review capacity and preserves selections under a strictly increasing within-day recalibration; precision and recall can still change under distribution shift. Exact ties use a seeded random draw. An earlier order-based tie-break let a constant score select late-day flights and was corrected.
- **Secondary: frozen threshold.** Choose the lowest threshold achieving empirical precision at least 0.40 on eligible Sep–Oct labels and freeze it before November. The precision target is not guaranteed on future data. A target of 0.50 was tried first and was unreachable on the calibration window.

D. Calibration rules

- **Static rule (fixed before test).** Fit Platt and isotonic on September outcomes available before the first October prediction; compare eligible October labels available before November. Keep raw unless the best recalibrator lowers October ECE by more than 0.005, then refit using eligible Sep–Oct labels before November.
- **Rolling Platt (adaptive alternative).** For each day D, refit Platt on outcomes *available* by D's decision time over the trailing N days and apply it to D's flights. It uses the same

availability dates as the features and is causal. $N = 28$ was fixed a priori; $N = 14$ and $N = 56$ are sensitivity only.

E. Uncertainty

- **Day-block bootstrap** (500 replicates) [23]. Flights on one day share weather, air-traffic programmes and cascading delays, so their errors are correlated; resampling flights would treat about 885 correlated flights per day as independent and give intervals far too narrow.
- **Paired differences.** Model differences are computed on the same resampled days, which removes the shared day-to-day variation.

F. Development protocol and disclosure

- **The test window was not touched only once.** Test metrics were visible in: (1) the first single-September design; (2) a month-feature comparison, the original and now withdrawn justification for omitting month; (3) feature-subset diagnostics that shaped the day-level hypothesis; (4) lowering the precision target from 0.50 to 0.40, decided because 0.50 is unreachable on Sep–Oct, yet test recall was printed; (5) before any calibration rule existed; (6) one pipeline run where the final-model comparison still used XGBoost's optimistic in-fold score, corrected for a methodological reason found from CV alone; (7) when adding the prior to the policy table exposed the order-based tie-break.
- **Fitting-clock correction.** A later audit found that chronological membership alone admitted some labels unavailable at fitting and selection boundaries. The reproduced run applies availability masks throughout and re-executes the pipeline and notebooks. Its updated results were also observed; the correction is methodological and does not create a pristine test set retroactively.
- **Evidence is therefore separated.** Everything used to *decide* (clock, m, window, C, XGBoost configuration, final model, calibration rule and its outcome, month removal, ablation claims) comes from **CV or Sep–Oct**. The rolling-calibration window ($N = 28$), the 10% share and the precision target were set a priori or on Sep–Oct. **Test results are exploratory** and small differences should be read with caution.

VIII. Evidence Used for Selection (CV and Sep–Oct)

A. Final-model choice

- **Fixed-tree comparison.** XGBoost's search score (0.4761) early-stops on the scored validation folds and is optimistic. Its fixed, fold-averaged tree count scores **0.4474**; LogReg scores **0.4715**. Route history scores 0.3477 and the prior 0.2777 (Table IX). Configurations and tree count still originate from the same CV folds: these are development scores, not independent nested-CV estimates.
- **Per-fold differences** (LogReg − XGBoost): $+0.001$, $+0.086$, $+0.006$, $+0.003$; mean $+0.024$. LogReg leads in all four folds; the paired-difference sd is 0.041 and each family's fold sd is about 0.07. Four folds do not establish population-level superiority.
- **Conclusion.** LogReg is chosen by the declared higher-CV-mean rule. The test bootstrap CI includes zero; this fails to establish a test-performance difference and does not prove equivalence.
- **A disclosed reversal.** Choosing with the optimistic in-fold score would favour XGBoost (0.4761 against 0.4715), which is why that score is excluded from family selection.

B. Calibration rule outcome

- **LogReg → Platt.** Eligible October ECE improves by 0.030, above the 0.005 margin. Raw LogReg over-predicts on this calmer calibration window.
- **XGBoost → raw.** The best recalibrator worsens eligible October ECE (gain -0.022), so the rule keeps raw scores.
- **Names.** "LogReg (Platt, rule-selected)" and "XGBoost (raw, rule-selected)" denote these variants throughout (Table X). Sep–Oct is also where the secondary policy's threshold was frozen.

C. Feature families

- **LogReg gains in CV.** Congestion $+0.020$ (4/4 folds), entity history $+0.020$ (4/4), day-level history $+0.011$ (3/4) (Fig. 4). These checks support retaining the families at the selected C; subsets are not separately retuned.
- **XGBoost.** Congestion gains 0.010 on average (3/4 folds); entity history and day-level history average slightly negative gains (respectively -0.0004 and -0.0007). The aggregate can be negative despite most fold gains being positive when a loss in one fold is larger. These subset checks keep the selected XGBoost configuration fixed.
- **Complementarity.** History alone (LogReg, 25 features) scores 0.4480 in CV, below the full feature set's 0.4715.
- **Grouped permutation importance** (Sep–Oct; Table XXIV, Fig. 5) shuffles each family together. LogReg AP drops: congestion 0.047, schedule 0.042, identifiers 0.038, entity history 0.029, day-level history 0.026. For XGBoost, schedule has the largest drop (0.056). This is descriptive; permutation breaks feature relationships and does not identify causal contributions.

IX. Exploratory Test-Set Findings (Nov–Dec)

The current run applies the declared selection rules before scoring the test window. Because test results were seen during development and later methodological corrections, this section is exploratory rather than an untouched final assessment.

A. Headline metrics

- **Results.** The final model, LogReg (Platt, rule-selected): PR-AUC **0.394**, ROC-AUC 0.654. XGBoost (raw, rule-selected): 0.404 and 0.662. Route history: 0.353 and 0.621. Prior: 0.258 and 0.500 (Table XI, Fig. 6). Both learned models have higher observed AP than the references.

TABLE IX
ROLLING-CV PR-AUC (VALIDATION MONTHS MAY–AUG), MODELS SCORED AS DEPLOYED.

Model	Calibration	CV PR-AUC	CV sd	May	Jun	Jul	Aug
Prior (base rate)	none	0.278	0.048	0.227	0.315	0.322	0.247
Route history	none	0.348	0.060	0.277	0.391	0.403	0.320
LogReg (as deployed)	raw	0.472	0.071	0.435	0.505	0.552	0.393
XGBoost (as deployed, fixed trees)	raw	0.447	0.069	0.434	0.419	0.547	0.390
XGBoost (in-fold early stopping, optimistic; not used)	raw	0.476	0.070	0.440	0.505	0.559	0.400

TABLE X
CALIBRATION RULE: FIT ON SEPTEMBER, EVALUATED ON OCTOBER.

Model	Calibrator	Oct ECE	Oct ECE (eq.-mass)	Oct Brier	ECE gain vs raw
logreg	raw	0.063	0.063	0.140	0.000
logreg	platt	0.033	0.033	0.137	0.030
logreg	isotonic	0.034	0.034	0.138	0.028
xgboost	raw	0.020	0.022	0.137	0.000
xgboost	platt	0.042	0.042	0.139	-0.022
xgboost	isotonic	0.042	0.042	0.140	-0.022

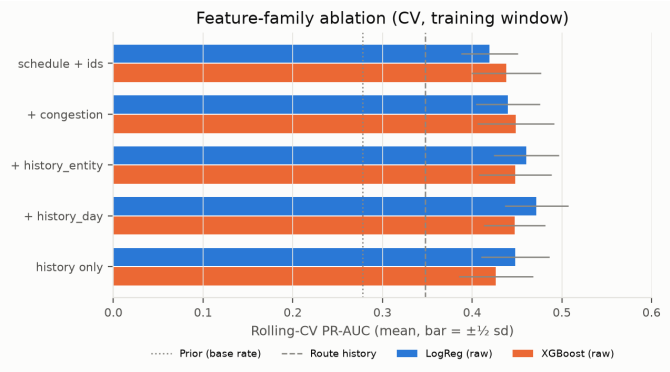


Fig. 4. Rolling-CV PR-AUC of cumulative feature sets, with prior (dotted) and route-history (dashed) references.

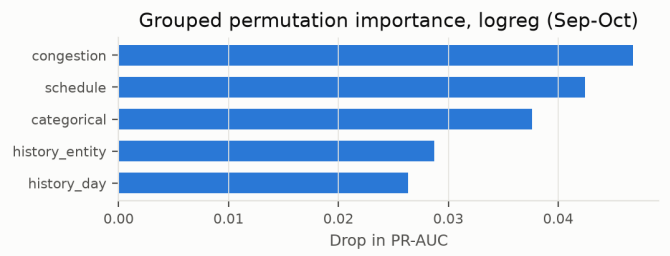


Fig. 5. Grouped permutation importance of the final model.

- **Platt is monotone.** LogReg (Platt) and LogReg (raw) rank identically; they differ in calibration (ECE 0.082 against 0.019), which is the subject of the calibration subsection below.
- **The prior's low ECE (0.007) is marginal calibration.** Both ECE variants use one bin for constant scores and equal $|\bar{p} - \bar{y}|$, which rounds to 0.007. AP equals the test base rate and ROC-AUC is 0.5, so this prior provides no flight-specific discrimination.

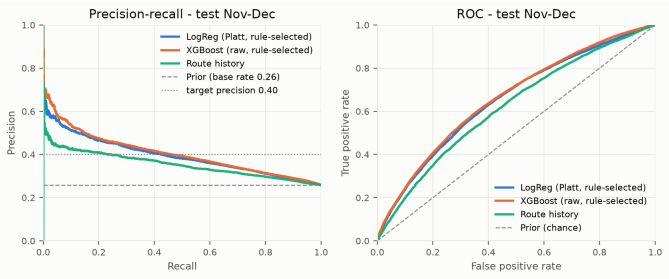


Fig. 6. Precision–recall and ROC curves on test. Dashed grey: prior.

- **Day-block bootstrap** (Table XII). XGBoost minus LogReg is $+0.010$ AP with 95% CI $[-0.013, +0.037]$; the ROC-AUC interval is $[-0.013, +0.027]$. Both include zero; neither establishes a difference or equivalence.
- **By month** (Table XXIII, appendix). The test window is split into November and December, because the base rate differs strongly (18.1% vs 33.5%).

B. *Operating policies*

- **Primary (daily top 10%), LogReg.** Precision **0.445**, recall 0.173, lift **1.72**. XGBoost lift is 1.67; route history 1.39; the seeded random prior ranking achieves 0.94 in this realisation (Table XIII).
- **Capacity and precision.** Calibration-window precision is 0.303 (lift 1.93), compared with 0.445 (lift 1.72) on test. Absolute precision rises while lift falls; fixed daily capacity does not imply invariant ranking quality.
- **Secondary (frozen threshold).** The LogReg threshold 0.330 targets calibration precision at least 0.40. Test precision is 0.495 and recall 0.142, flagging 7.4% of flights; calibration flags 3.8%. The review load changes across windows, even though the empirical precision target is met on this test set. The prior never meets the calibration precision target and flags nothing.

TABLE XI
TEST METRICS (NOV–DEC). ORACLE REC. IS THE BEST RECALL AT PRECISION ≥ 0.40 WITH THE THRESHOLD TUNED ON TEST LABELS: COMPARISON ONLY.

Model	Calibration	PR-AUC	ROC-AUC	Brier	ECE	ECE (eq.-mass)	Oracle rec.@0.40
Prior (base rate)	none	0.258	0.500	0.192	0.007	0.007	0.000
Route history	none (smoothed rate)	0.353	0.621	0.187	0.037	0.038	0.244
LogReg (raw)	raw	0.394	0.654	0.181	0.019	0.022	0.431
LogReg (Platt, rule-selected)	Platt (rule-selected)	0.394	0.654	0.187	0.082	0.082	0.431
LogReg (rolling Platt 28d)	rolling Platt 28d (causal)	0.392	0.657	0.181	0.027	0.027	0.431
XGBoost (raw, rule-selected)	raw (rule-selected)	0.404	0.662	0.180	0.027	0.026	0.462
XGBoost (rolling Platt 28d)	rolling Platt 28d (causal)	0.394	0.659	0.181	0.026	0.025	0.417

TABLE XII
DAY-BLOCK BOOTSTRAP ON TEST (500 REPLICATES).

Metric	XGBoost 95% CI	LogReg 95% CI	XGB−LR 95% CI	P(XGB not better)
PR-AUC	[0.346, 0.466]	[0.34, 0.45]	[-0.013, 0.037]	0.236
ROC-AUC	[0.631, 0.692]	[0.62, 0.682]	[-0.013, 0.027]	0.204

C. Calibration under seasonal shift

- **Static Platt.** Fitted on eligible calm Sep–Oct outcomes, it scales risk *downward*: November ECE is 0.022, but December ECE is **0.143** at a base rate of 33.5%. Isotonic behaves similarly. **The static rule fails in December.**
- **Raw LogReg** has low overall ECE (0.019), but November and December ECE are 0.056 and 0.059. The aggregate masks opposite monthly probability errors; it does not establish calibration across seasons.
- **Rolling Platt** ($N = 28$) improves probability quality in this run: test ECE 0.027 for LogReg (November 0.006, December 0.050) and 0.026 for XGBoost. LogReg sensitivity values are 0.022 at $N = 14$ and 0.051 at $N = 56$; these are exploratory comparisons, while $N = 28$ was declared in advance.
- **Shape versus level.** Raw predictions follow the *shape* of the season but compress it toward the middle (Fig. 8).
- **Interpretation.** The result motivates monitoring calibration under shift [3] and testing updates on recent *known* outcomes. Rolling recalibration improves this run but is not guaranteed to succeed under every shift (Table XIV, Fig. 7).

D. Ablation on test

- **LogReg.** Entity history has a positive exploratory difference ($+0.032$, CI $[+0.015, +0.049]$); congestion ($+0.007$) and day-level history ($+0.009$) intervals include zero. These are subset diagnostics at the selected C, not independent confirmations.
- **XGBoost.** Entity history shows $+0.055$ on test although its mean CV gain is slightly negative: an unresolved difference across periods, with hyperparameters held fixed (Table XV).

E. Target definition

- **Cancelled and diverted counted as delayed.** The test base rate rises from 25.8% to 27.7% and PR-AUC with it (LogReg 0.446, XGBoost 0.455; the prior moves in step).

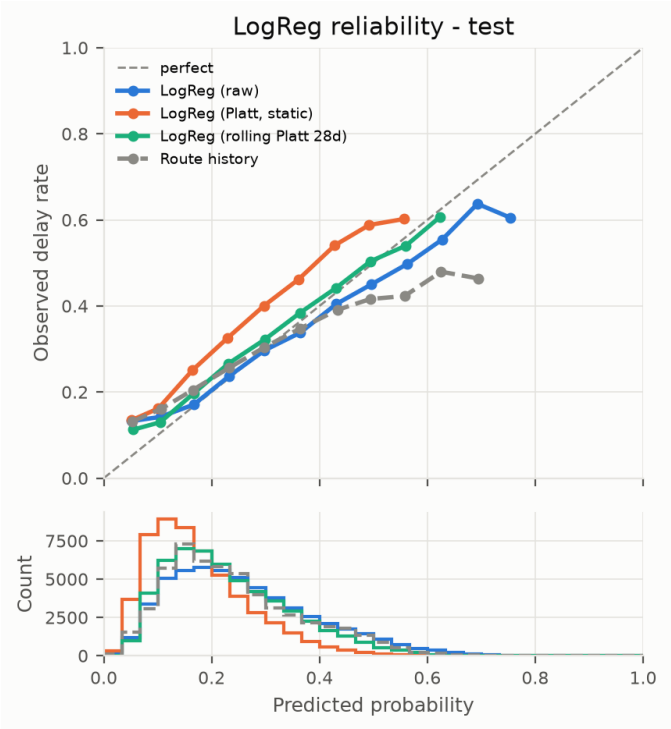


Fig. 7. Reliability of logistic regression on test: raw, static Platt, rolling Platt, route history.

- **Conclusions hold.** The XGBoost minus LogReg CI $[-0.013, +0.037]$ includes zero; the daily top-10% lift is 1.73 (LogReg), 1.71 (XGBoost) and 0.95 (prior); both learned models still beat both references (Table XVI).

F. Covariate shift

- **Methods.** Population stability index (PSI; training as reference; 0.25 is a heuristic warning level) and adversarial validation (a classifier separating training from later rows)

TABLE XIII
OPERATING POLICIES ON THE TEST WINDOW.

Model	Policy	Flag rate	Precision	Recall	Lift	Threshold
Prior (base rate)	daily top 10%	0.101	0.243	0.095	0.941	–
Prior (base rate)	frozen thr. (P≥0.4)	0.000	–	0.000	–	∞
Route history	daily top 10%	0.101	0.358	0.139	1.387	–
Route history	frozen thr. (P≥0.4)	0.059	0.425	0.097	1.646	0.469
LogReg (Platt, rule-selected)	daily top 10%	0.101	0.445	0.173	1.722	–
LogReg (Platt, rule-selected)	frozen thr. (P≥0.4)	0.074	0.495	0.142	1.917	0.330
XGBoost (raw, rule-selected)	daily top 10%	0.101	0.431	0.168	1.667	–
XGBoost (raw, rule-selected)	frozen thr. (P≥0.4)	0.088	0.489	0.167	1.892	0.417

TABLE XIV
CALIBRATION OF EVERY VARIANT ON TEST, NOVEMBER AND DECEMBER.

Model	ECE test	ECE Nov	ECE Dec	Brier test
Prior (base rate)	0.007	0.084	0.070	0.192
Route history	0.037	0.035	0.060	0.187
LogReg (raw)	0.019	0.056	0.059	0.181
LogReg (Platt, static)	0.082	0.022	0.143	0.187
LogReg (isotonic, static)	0.082	0.023	0.142	0.187
LogReg (rolling Platt 14d)	0.022	0.011	0.038	0.181
LogReg (rolling Platt 28d)	0.027	0.006	0.050	0.181
LogReg (rolling Platt 56d)	0.051	0.014	0.089	0.182
XGBoost (raw)	0.027	0.017	0.061	0.180
XGBoost (Platt, static)	0.073	0.031	0.117	0.185
XGBoost (isotonic, static)	0.071	0.033	0.113	0.185
XGBoost (rolling Platt 14d)	0.028	0.013	0.050	0.180
XGBoost (rolling Platt 28d)	0.026	0.011	0.042	0.181
XGBoost (rolling Platt 56d)	0.036	0.021	0.053	0.180

TABLE XV
ABLATION STEPS: CV GAIN (SELECTION EVIDENCE) AND TEST DIFFERENCE WITH 95% CI (EXPLORATORY).

Model	Calibration	Step	CV gain	Folds up	Test diff.	CI low	CI high
LogReg	raw	+ congestion	0.020	4/4	0.007	-0.009	0.028
LogReg	raw	+ history_entity	0.020	4/4	0.032	0.015	0.049
LogReg	raw	+ history_day	0.011	3/4	0.009	-0.009	0.027
XGBoost	raw	+ congestion	0.010	3/4	-0.003	-0.017	0.013
XGBoost	raw	+ history_entity	-0.000	2/4	0.055	0.030	0.076
XGBoost	raw	+ history_day	-0.001	3/4	0.005	-0.029	0.043

TABLE XVI
TARGET SENSITIVITY: CANCELLED AND DIVERTED FLIGHTS COUNTED AS DELAYED.

Target	Model	Base rate	PR-AUC	ROC-AUC	Top-10% lift
arr_delay≥15 (main)	Prior (base rate)	0.258	0.258	0.500	0.941
arr_delay≥15 (main)	Route history	0.258	0.353	0.621	1.387
arr_delay≥15 (main)	LogReg (raw)	0.258	0.394	0.654	1.722
arr_delay≥15 (main)	XGBoost (raw)	0.258	0.404	0.662	1.667
arr_delay≥15 OR cancelled/diverted	Prior (base rate)	0.277	0.277	0.500	0.946
arr_delay≥15 OR cancelled/diverted	Route history	0.277	0.374	0.621	1.359
arr_delay≥15 OR cancelled/diverted	LogReg (raw)	0.277	0.446	0.661	1.729
arr_delay≥15 OR cancelled/diverted	XGBoost (raw)	0.277	0.455	0.672	1.708

[15] describe distribution change. They do not prove its causes or directly establish calibration failure.

- **Adversarial validation** (Table XXV). All features together separate training from later periods perfectly (AUC 1.00). By family: identifiers nearly stable (0.56), schedule moderately (0.78 against Sep–Oct, 0.92 against test), congestion strongly (0.98, 0.96) because daily volumes differ by season, and history features completely, as designed.
- **PSI** (Table XXVI, Fig. 9). `days_to_holiday` shifts most into Nov–Dec (4.8) because Thanksgiving, Christmas and New Year fall in the test window; the history rates move most into Sep–Oct.
- **Implication.** Some day-level features are shared across an entire date and strongly identify the period. This raises a

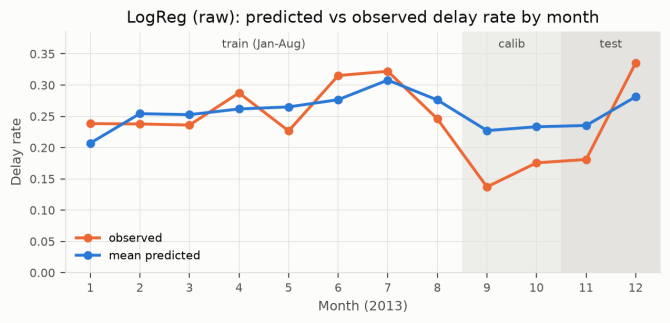


Fig. 8. Predicted vs observed delay rate by month (raw logistic regression).

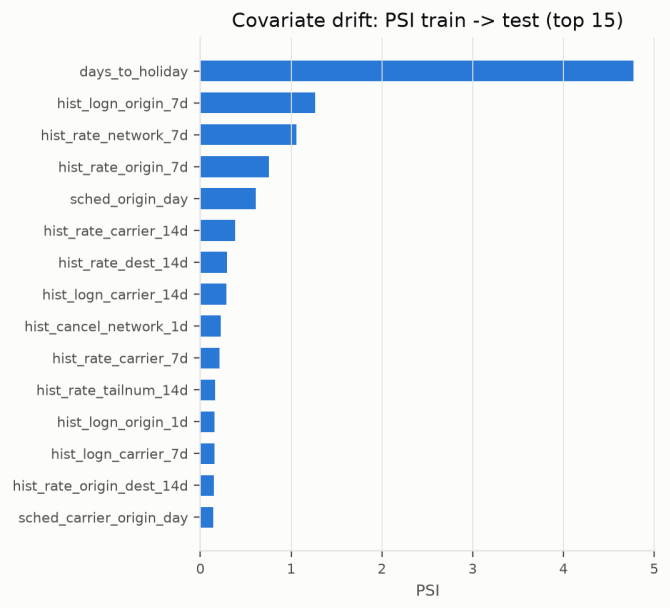


Fig. 9. PSI, training against the test window (top 15 features).

memorisation concern for flexible models; period separability alone does not show that such memorisation occurred or caused the calibration errors.

X. Decision-Support Analysis: Rescheduling Candidates

The brief requests an offline what-if rescheduling analysis under explicit assumptions. The implemented Phase 1 analysis supplies its candidate-selection component: whether daily risk ranking concentrates delayed flights for review. It does not alter schedules or estimate the resulting delays. A full what-if simulation remains an outstanding deliverable, with feasibility and intervention effects stated separately below.

A. Design

- **Setup.** Each test day, the final model's ranking selects the top-$k\%$ flights as **reschedule / add-buffer candidates**, $k \in \{5, 10, 15, 20\}$. The test window has 61 days and about 885 flights per day, so $k = 5\%$ is about 45 flights and $k = 20\%$ about 177 flights per day.
- **Compared with.** The prior (random order) and route-history rankings, plus XGBoost.
- **Reported.** Flagged per day, precision, recall, lift, and delayed flights per 100 flagged (Table XVII).
- **Fixed in advance.** The k grid and the policy were specified before the analysis ran; there is no tuning on test.

B. Results

- **Top 5%** ($\approx$45 flights/day). LogReg precision **0.470**, lift 1.82, i.e. **47 delayed per 100 flagged**; about 21 delayed flights captured per day, against 17 for route history and 11 for the random prior ranking.
- **Top 10%.** Precision 0.445, recall 0.173, lift 1.72 (identical to the primary policy of Sec. IX); about 40 delayed flights captured per day against about 32 for route history and 22 for the prior ranking.
- **Top 20%** ($\approx$177 flights/day). Precision 0.406, lift 1.57, recall 0.315, about 72 delayed flights caught per day (route history about 60, random about 45).
- **The trade-off.** Widening the desk from 5% to 20% catches more delays but with lower precision: the extra $\approx$133 flights per day are late about 38% of the time, against 47% for the top 5%. The ranking therefore degrades smoothly, with no cliff.
- **Ordering holds at every k.** LogReg $\geq$ XGBoost $>$ route history $>$ prior (Table XVII); LogReg versus XGBoost is not a distinguished difference (Sec. IX). Random flagging yields about 25 per 100 at every k, i.e. the base rate, as it should.

C. How to read it

- **Capacity as the design variable.** A desk that can review only 5% of flights should expect about 47 true delays per 100 reviews; one that can review 20% should expect about 41. The marginal value of added capacity falls, which is useful when staffing the desk.
- **Why rank-based.** A daily share maintains the specified review capacity. The positive-slope static Platt mapping preserves LogReg's ordering, so its December calibration error does not change these selections; their precision and recall can still deteriorate as conditions change.
- **Failure modes.** The ranking cannot know which delays are *preventable*; flights delayed by weather or air-traffic programmes are flagged but cannot be saved by rescheduling. Recall at 10% is only 0.173, so most delayed flights are never flagged.

D. What this shows — and does not

- **Shows: ranking quality** for a capacity-limited review desk.
- **Does not show: realised operational savings**. There is no cost model for delay or for a schedule change, so the benefit of moving or buffering a flight cannot be estimated.
- **Missing for a full what-if study:** an explicit recommendation rule, feasible alternative schedules (gates, crews, slots and rotations), an assumed effect of each intervention, and

TABLE XVII
RESCHEDULING-CANDIDATE ANALYSIS ON THE TEST WINDOW: DAILY TOP-$k\%$ BY RISK.

Model	Top share	Flagged/day	Precision	Recall	Lift	Delayed per 100
Prior (base rate)	5%	44.705	0.251	0.049	0.971	25.083
Prior (base rate)	10%	88.984	0.243	0.095	0.941	24.300
Prior (base rate)	15%	133.148	0.250	0.145	0.967	24.982
Prior (base rate)	20%	177.393	0.252	0.195	0.974	25.164
Route history	5%	44.705	0.386	0.076	1.495	38.614
Route history	10%	88.984	0.358	0.139	1.387	35.814
Route history	15%	133.148	0.346	0.201	1.339	34.597
Route history	20%	177.393	0.339	0.263	1.313	33.916
LogReg (Platt, rule-selected)	5%	44.705	0.470	0.092	1.819	46.975
LogReg (Platt, rule-selected)	10%	88.984	0.445	0.173	1.722	44.473
LogReg (Platt, rule-selected)	15%	133.148	0.422	0.246	1.635	42.243
LogReg (Platt, rule-selected)	20%	177.393	0.406	0.315	1.571	40.579
XGBoost (raw, rule-selected)	5%	44.705	0.466	0.091	1.803	46.571
XGBoost (raw, rule-selected)	10%	88.984	0.431	0.168	1.667	43.055
XGBoost (raw, rule-selected)	15%	133.148	0.413	0.240	1.599	41.295
XGBoost (raw, rule-selected)	20%	177.393	0.399	0.310	1.546	39.922

comparison of predicted risk or delay cost before and after the proposed change. The current dataset does not establish causal intervention effects; any later simulated benefit must be labelled assumption-dependent.

XI. DISCUSSION AND LIMITATIONS

A. What the evidence supports

- **The ranking supports candidate review.** LogReg's top tenth has $1.72\times$ base-rate precision, against $1.39\times$ for route history. This demonstrates concentration of observed delays, not prevention or financial benefit.
- **Representation remains a hypothesis.** The learned-model difference is uncertain on test. This does not establish whether manual summaries exhaust the usable interaction structure; later embedding MLP experiments should test that question fairly.
- **Feature families help LogReg in these CV checks.** Congestion, entity and day-level history increase mean AP at the selected C; entity history also has a positive exploratory test difference.
- **Calibration needs monitoring.** Static Platt fails in December. Ranking fixes review capacity, while frozen probability thresholds have a changing load; neither policy guarantees stable quality under shift.

B. Negative and unresolved results

- **The static calibration rule fails** in December (ECE 0.143). It is reported as is.
- **Equal-mass ECE** is unstable for tied predictions in the current implementation. The fixed-width ECE and Brier discussion remains separate from this unresolved diagnostic issue.
- **XGBoost's ablation** disagrees between CV and test for entity history.
- **The final-model choice** would have reversed under the optimistic in-fold score.
- **The original month-feature and day-level justifications** relied on test numbers and were withdrawn and replaced by CV evidence.

C. Threats to validity

- **Internal.** The test window was seen during development (Sec. VII); we limit this by deciding only from CV and Sep–Oct, but the separation cannot be perfect. Four CV folds cannot separate models differing by thousandths.
- **External.** One year and three origins: each season and airport effect is observed once, and December is a single draw. Cross-year and cross-airport generalisation cannot be established.
- **Construct.** The label is a 15-minute threshold; the sensitivity analysis shows conclusions survive counting cancellations as delays, but other thresholds (30, 60 minutes) were not modelled.
- **Statistical.** Day-block intervals account for within-day correlation but not for correlation across neighbouring days; with 61 test days they are wide.

D. Limitations and assumptions

- **No weather, by design.** Observed weather for D would leak at the clock; only archived forecasts issued before 20:00 could be added. Weather is the main missing driver of day-to-day delay, and this is a stated gap rather than a defect.
- **Partial rotation.** Only NYC departures are visible, about 1.3 per aircraft per day.
- **Assumptions.** Tail-number assignment is known by 20:00 on $D-1$; holidays follow the US federal calendar; a cancellation is known at the scheduled departure and an arrival at the actual gate arrival; destination time zones come from a fixed table.

E. Future work

- Add archived weather *forecasts* issued before the clock.

- A compact MLP with learned airline, origin and destination embeddings, evaluated against the same ML features, clock and date splits. A temporal model is optional only after constructing correctly ordered histories from available past outcomes.
- Multi-year data to test cross-year generalisation of the calibration findings.
- An offline what-if recommendation rule with feasible schedule alternatives, explicit assumptions about intervention effects, and before/after risk or cost comparisons.

XII. Conclusion

- **Clock first.** The 20:00 $D-1$ cutoff governs predictors, training labels, CV selection, calibration and thresholds. Moving the clock to midnight raises CV AP by about 0.010; feature and fitting-boundary regression tests guard the stated protocol.
- **Models.** Logistic regression is the declared final model (CV PR-AUC 0.4715 against fixed-tree XGBoost 0.4474). The test bootstrap fails to establish a model difference; it does not prove equivalence.
- **Policy.** The daily top 10% reaches precision 0.445 and lift 1.72 in this dataset. Rolling recalibration improves ECE to 0.027 compared with static Platt 0.082 overall and 0.143 in December. Candidate ranking does not demonstrate delay reductions.
- **Honesty.** Selection evidence (CV, Sep–Oct) is kept apart from exploratory test findings, and every test exposure is disclosed.
- **Next.** Later work will test the brief's embedding MLP, optional ordered-history temporal model, archived forecasts and an explicit offline what-if recommendation rule.

Reproducibility

Python 3.12 and dependencies pinned in `requirements.txt` are used in a project-local `.venv`; macOS XGBoost also needs OpenMP. Download the Kaggle CSV to `data/raw/flights.csv`, which is excluded from Git, and register the `aerocast` kernel. From the repository root, `python -B scripts/run_phase1.py` regenerates `results/` and `figures/`. Execute the four notebooks using `python -B -m nbconvert`; notebooks 03 and 04 check their tables against saved results. The full MacBook Pro pipeline run on 23 October 2026 took about 22 minutes; runtime varies. The expanded suite passes 42 tests; ECE was updated from saved scores without classifier retraining. All randomness is seeded (42). Report assets are exported with `scripts/export_report_assets.py`; `scripts/check_report_numbers.py` checks quoted numeric strings, but does not validate their surrounding interpretation. PDF compilation and visual review are separate checks. The full `scripts/reproduce.sh` additionally cleans generated/report assets and attempts a PDF build if a LaTeX engine is installed. Table XXVIII lists the configuration. Changing datasets requires reviewing the date splits, calendar/time-zone assumptions and dataset-specific assertions, not just matching columns.

TABLE XVIII
Repository layout and reproducibility responsibilities.

Directory	Role
`data/raw/`	Local raw CSV; excluded from Git
`src/aerocast/`	Shared data, features, models, evaluation and pipeline logic
`scripts/`	Pipeline runner, report export and numeric checks
`notebooks/`	Four executed stages importing shared logic
`tests/`	Leakage, availability, time-zone, metric and export checks
`results/`	Saved CSV/JSON tables and prediction audit trail
`figures/`	Generated PNG plots
`docs/`	Stage explanations, protocol and limitations
`report/`	IEEE source, bibliography, PDF and exported assets

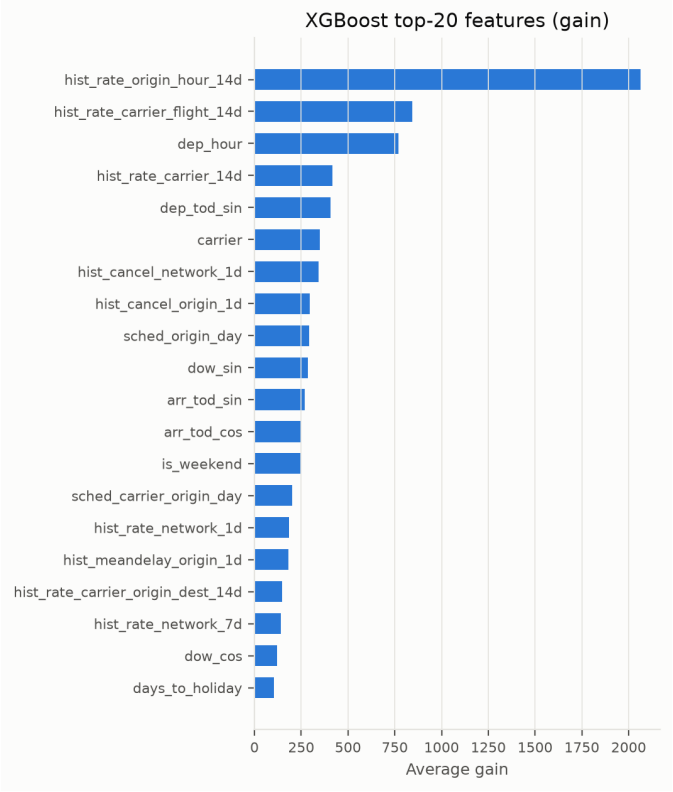


Fig. 10. XGBoost top-20 features by gain.

Appendix A
Additional Tables and Figures

Supporting material: full ablation (Table XXVII), test metrics by month (Table XXIII), grouped permutation importance (Table XXIV), XGBoost gain (Fig. 10), XGBoost reliability (Fig. 11), the C curve (Fig. 12) and why a single validation month was abandoned (Fig. 13).

TABLE XIX
HISTORY SETTINGS TUNED JOINTLY WITH C (ROLLING-CV PR-AUC OF LOGREG). CLOCK 20 IS DEPLOYED; THE LAST ROWS MOVE THE CLOCK TO 00:00 AND 24:00.

m	Window (d)	Clock (h)	Best C	CV PR-AUC	CV sd	CV (C=0.001)
5	14	20	0.100	0.472	0.071	0.469
5	28	20	0.001	0.467	0.064	0.467
5	56	20	0.010	0.469	0.069	0.467
20	14	20	0.100	0.470	0.071	0.468
20	28	20	0.001	0.466	0.064	0.466
20	56	20	0.010	0.468	0.069	0.466
50	14	20	0.010	0.466	0.072	0.464
50	28	20	0.001	0.464	0.064	0.464
50	56	20	0.010	0.465	0.069	0.463
5	14	0	0.003	0.459	0.074	0.458
5	14	24	0.010	0.481	0.072	0.480

TABLE XX
ROLLING-CV PR-AUC WITHOUT AND WITH CYCLIC MONTH FEATURES.

Features	Model	May	Jun	Jul	Aug	CV PR-AUC
without month	logreg	0.435	0.505	0.552	0.393	0.472
without month	xgboost	0.434	0.419	0.547	0.390	0.447
with month	logreg	0.426	0.507	0.549	0.395	0.469
with month	xgboost	0.374	0.447	0.552	0.389	0.440

TABLE XXI
LOGREG: ROLLING-CV PR-AUC BY C (MEAN, SD OVER FOUR FOLDS).

C	CV PR-AUC	CV sd
0.000	0.466	0.069
0.001	0.469	0.070
0.003	0.471	0.070
0.010	0.471	0.071
0.100	0.472	0.071
1.000	0.471	0.071

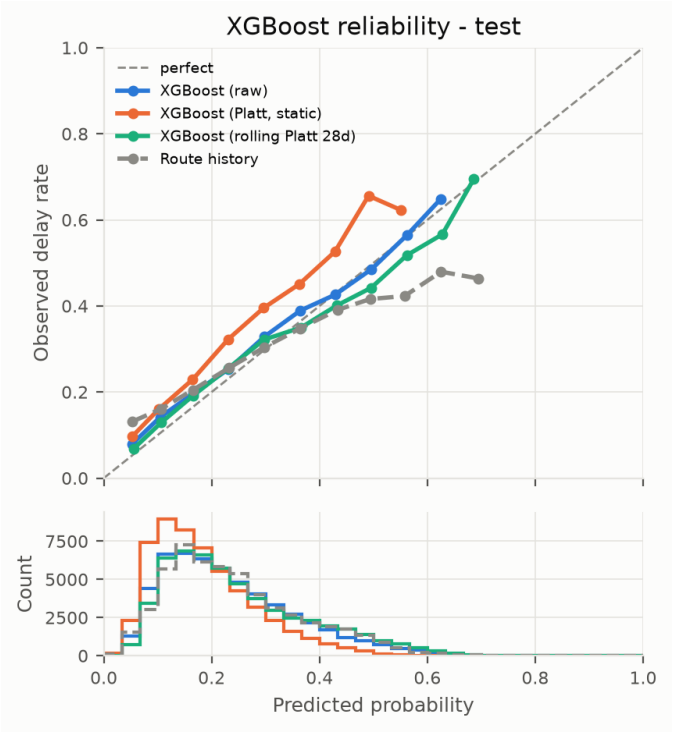


Fig. 11. Reliability of XGBoost on test: raw, static Platt, rolling Platt, route history.

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TABLE XXII
CHRONOLOGICAL SPLIT (CANCELLED AND DIVERTED FLIGHTS DROPPED).

Split	Flights	Delay rate	From	To
train	211684	0.265	2013-01-08	2013-08-31
calib	55628	0.157	2013-09-01	2013-10-31
test	53991	0.258	2013-11-01	2013-12-31

TABLE XXIII
TEST METRICS BY MONTH.

Model	Split	PR-AUC	ROC-AUC	Brier	ECE
Prior (base rate)	Nov	0.181	0.500	0.155	0.084
Route history	Nov	0.221	0.560	0.149	0.035
LogReg (raw)	Nov	0.292	0.655	0.145	0.056
LogReg (Platt, rule-selected)	Nov	0.292	0.655	0.142	0.022
LogReg (rolling Platt 28d)	Nov	0.286	0.650	0.142	0.006
XGBoost (raw, rule-selected)	Nov	0.267	0.634	0.144	0.017
XGBoost (rolling Platt 28d)	Nov	0.259	0.627	0.144	0.011
Prior (base rate)	Dec	0.335	0.500	0.228	0.070
Route history	Dec	0.394	0.574	0.225	0.060
LogReg (raw)	Dec	0.456	0.632	0.217	0.059
LogReg (Platt, rule-selected)	Dec	0.456	0.632	0.233	0.143
LogReg (rolling Platt 28d)	Dec	0.432	0.603	0.220	0.050
XGBoost (raw, rule-selected)	Dec	0.463	0.634	0.216	0.061
XGBoost (rolling Platt 28d)	Dec	0.438	0.612	0.217	0.042

TABLE XXIV
GROUPED PERMUTATION IMPORTANCE: PR-AUC DROP ON SEP–OCT.

Group	LogReg	XGBoost
categorical	0.038	0.015
congestion	0.047	0.011
history_day	0.026	0.017
history_entity	0.029	0.019
schedule	0.042	0.056

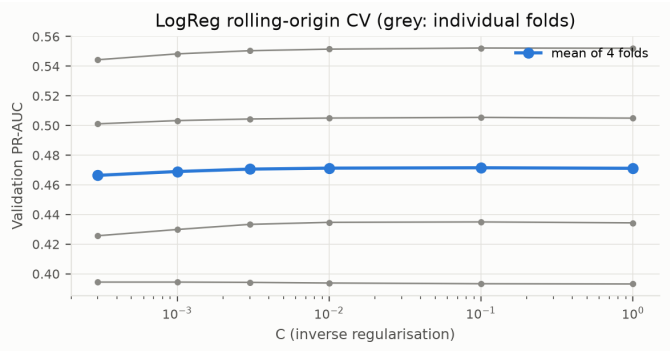


Fig. 12. Logistic regression rolling-CV PR-AUC against C.

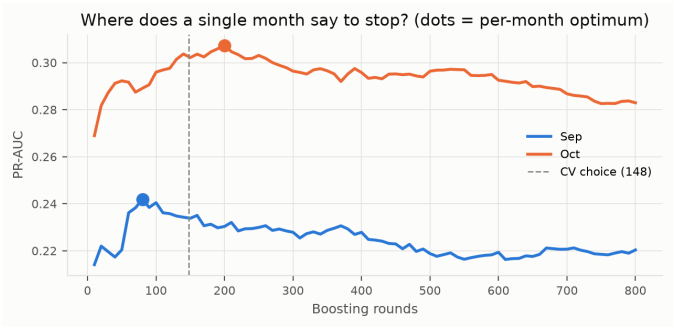


Fig. 13. Why a single validation month was abandoned: XGBoost PR-AUC on September and October vs number of trees.

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TABLE XXV
ADVERSARIAL VALIDATION: AUC OF A CLASSIFIER SEPARATING TRAINING ROWS FROM LATER ROWS.

Comparison	Features	#feat.	Adversarial AUC
train vs calib	all features	45	1.000
train vs calib	categorical	3	0.562
train vs calib	schedule	10	0.785
train vs calib	congestion	7	0.983
train vs calib	history_entity	18	1.000
train vs calib	history_day	7	1.000
train vs test	all features	45	1.000
train vs test	categorical	3	0.562
train vs test	schedule	10	0.921
train vs test	congestion	7	0.963
train vs test	history_entity	18	0.998
train vs test	history_day	7	1.000

TABLE XXVI
LARGEST POPULATION-STABILITY SHIFTS (TRAINING REFERENCE).

Feature	Group	PSI (Sep–Oct)	PSI (test)
days_to_holiday	schedule	0.261	4.769
hist_logn_origin_7d	history_entity	2.269	1.265
hist_rate_network_7d	history_day	5.147	1.058
hist_rate_origin_7d	history_entity	3.433	0.755
sched_origin_day	congestion	2.056	0.610
hist_rate_carrier_14d	history_entity	2.631	0.383
hist_rate_dest_14d	history_entity	2.143	0.292
hist_logn_carrier_14d	history_entity	0.472	0.289
hist_cancel_network_1d	history_day	1.229	0.222
hist_rate_carrier_7d	history_entity	1.699	0.209

TABLE XXVII
FULL ABLATION: CUMULATIVE FEATURE SETS FOR BOTH MODELS AND THE REFERENCES.

Model	Calibration	Features	#feat.	CV PR-AUC	CV sd	Test PR-AUC	Test ROC-AUC
Prior (base rate)	none	(baseline)	0	0.278	0.048	0.258	0.500
Route history	none (smoothed rate)	(baseline)	0	0.348	0.060	0.353	0.621
LogReg	raw	schedule + ids	13	0.419	0.063	0.345	0.612
LogReg	raw	+ congestion	20	0.440	0.071	0.352	0.614
LogReg	raw	+ history_entity	38	0.460	0.073	0.384	0.643
LogReg	raw	+ history_day	45	0.472	0.071	0.394	0.654
LogReg	raw	history only	25	0.448	0.076	0.390	0.657
XGBoost	raw	schedule + ids	13	0.438	0.077	0.346	0.621
XGBoost	raw	+ congestion	20	0.448	0.085	0.344	0.618
XGBoost	raw	+ history_entity	38	0.448	0.081	0.399	0.652
XGBoost	raw	+ history_day	45	0.447	0.069	0.404	0.662
XGBoost	raw	history only	25	0.427	0.083	0.375	0.647

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TABLE XXVIII
FINAL CONFIGURATION (ALL FIXED BY CV OR SEP–OCT, NONE BY TEST).

Item	Value
Decision clock	20:00 NYC on $D-1$
Target	arrival delay ≥ 15 min
Shrinkage m, window	5, 14 days
LogReg C	0.1
XGBoost	depth 2, lr 0.1, mcw 20, row subsample 0.8, colsample 0.5, $\lambda = 1$, 148 trees
Final model	LogReg, Platt (rule-selected)
Rolling calibration	Platt, $N = 28$ days
Primary policy	daily top 10%
Secondary policy	frozen threshold 0.330 targeting calibration precision ≥ 0.40
Seed	42